

STAGE 1 (Technical)

Priority Checklist — Mineral Intelligence Platform (Ghana MVP)

1. One-Page Architecture Napkin

Technical Sketch — Africa Mineral Intelligence Platform (Ghana MVP)

Core User Flow (1 sentence)

User searches or views a mining region on an interactive map → system aggregates and visualizes mineral, production, and mining intelligence data → user receives actionable insights, analytics, and downloadable reports.

Components Needed

Frontend

React

Next.js

Tailwind CSS

Mapbox GL / Leaflet

Chart.js or D3.js

Backend

Node.js + NestJS/Express

Python microservice for GIS/data processing

REST or GraphQL API

Data Store

PostgreSQL + PostGIS

Redis (cache)

Object storage for datasets/files

Third-Party APIs / Data Sources

OpenStreetMap

Mapbox

Geological survey datasets

Ghana Minerals Commission data

Commodity price APIs

Satellite imagery APIs (future phase)

Auth

JWT authentication

Role-based access control (Admin / Researcher / Public User)

Simple Architecture Flow

User Browser ↓ Frontend Web App (React + Maps) ↓ Backend API Gateway ↓ -----
----- | PostgreSQL/PostGIS Database | | GIS Processing Service | | Analytics Engine | -----
----- ↓ External Mining & GIS Data Sources

2. The 3 Riskiest Technical Assumptions

1. Reliable Mining Data Availability

Risk:

Mining data across Ghana and ECOWAS may be incomplete, inconsistent, outdated, or politically restricted.

Why it matters:

The platform's value depends heavily on trusted data quality.

Mitigation:

Start with open/public datasets

Build partnerships with Ghana Minerals Commission and universities

Use multi-source verification

2. GIS & Large Map Performance

Risk:

Rendering large mineral datasets and map layers may become slow on low-end devices and weak internet connections.

Why it matters:

Users in parts of Ghana and West Africa may have:

low bandwidth

unstable internet

older smartphones/laptops

Mitigation:

Lazy-load map layers

Compress geospatial data

Use caching/CDNs

Mobile-first optimization

3. Scaling Multi-Country Data Architecture

Risk:

ECOWAS expansion introduces:

different data standards

multiple languages

legal differences

inconsistent formats

Why it matters:

Poor architecture early can make regional scaling extremely expensive later.

Mitigation:

Modular country-based data ingestion

Standardized schema design

Country abstraction layer

3. Must-Have APIs & Data Sources

Critical APIs / Services

Service	Purpose
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Mapbox or OpenStreetMap	Interactive maps
PostGIS	Geospatial queries
Ghana Minerals Commission datasets	Mining licenses & mineral data
Commodity price APIs	Gold/lithium pricing
OpenWeather API (optional)	Environmental overlays
Satellite imagery APIs (future)	Monitoring & detection

Important Data Sources

Ghana

Minerals Commission Ghana

Geological Survey Authority

Ghana Statistical Service

Environmental Protection Agency Ghana

ECOWAS Expansion

ECOWAS mining databases

African Union mineral reports

World Bank mining datasets

USGS mineral resources data

4. Irreducible Infrastructure Cost Per User

MVP Estimate (100 Active Users)

Item	Estimated Monthly Cost
Cloud server	\$80 – \$150
Database hosting	\$40 – \$100
Map services/API calls	\$50 – \$200
Storage & backups	\$20 – \$50
Monitoring/security	\$20 – \$50

Estimated Base Infrastructure Cost

~\$210 – \$550/month

At 100 users:

Approximate Cost Per User

~\$2 – \$5.50 per active user/month

This can reduce significantly with:

OpenStreetMap

Self-hosted GIS stack

Data caching

5. Can the MVP Ship in < 6 Weeks?

YES — if the MVP scope is controlled.

Realistic 6-Week MVP Scope

Week 1

Architecture setup

UI wireframes

Database schema

Map integration

Week 2

Authentication

Dashboard structure

Mineral data ingestion

Week 3

GIS layers

Interactive map filters

Basic analytics

Week 4

Reporting/export

Admin panel

Data visualization

Week 5

Optimization

Mobile responsiveness

Testing

Week 6

Deployment

Bug fixing

Demo preparation

Conditions Required

You MUST avoid:

AI features

Satellite intelligence

Real-time drone integrations

Multi-country support

Advanced predictive analytics

Recommended MVP Features ONLY

Interactive mining map

Ghana mineral locations

Production dashboard

Search/filter system

Admin data upload panel

Basic analytics

User authentication

6. 3 Common Mistakes

1. Starting Development Before Stakeholder Interviews

Mistake:

Building features without validating real mining-industry needs.

Risk:

You create a technically impressive platform nobody actually pays for.

Solution:

Conduct at least:

mining company interviews

regulators interviews

researchers interviews

government stakeholder interviews

before full development.

2. Overengineering the First Architecture

Mistake:

Trying to build a continental-scale system from day one.

Risk:

Complexity kills MVP speed.

Solution:

Build:

Ghana only

Single database

Simple modular backend

Manual data ingestion initially

3. Ignoring Ghanaian Internet & Device Constraints

Mistake:

Heavy GIS dashboards optimized only for fast internet.

Risk:

Slow adoption and poor usability.

Solution:

Lightweight frontend

Mobile-first optimization

Data compression

Offline-friendly caching

7. Stage-Gate Criteria (Technical Pillar)

Required Before Full Build

Architecture napkin exists and reviewed

Status:

Achieved.

No single technical blocker is fatal

Status:

Currently true.

Main challenges are:

data acquisition

GIS optimization

scaling architecture

but none are impossible.